

TIEboard: Developing Kids Geometric Thinking through Tangible User Interface

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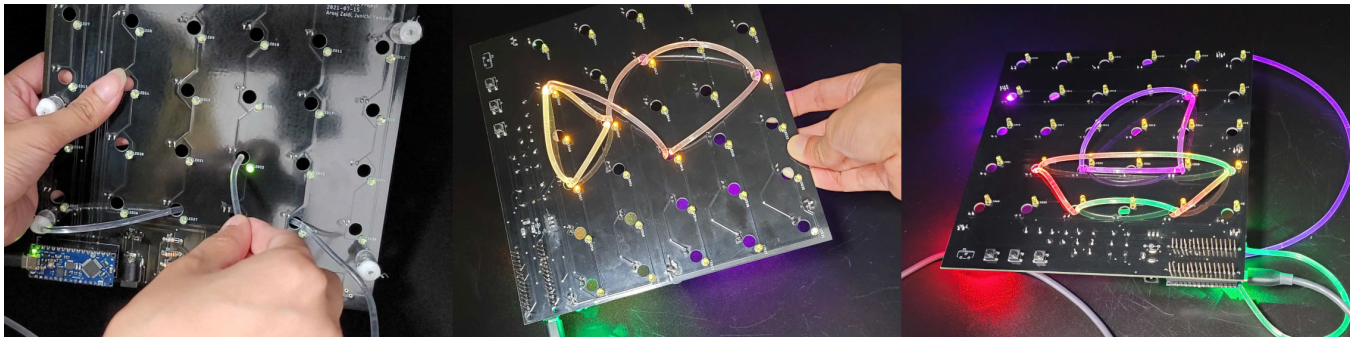


Figure 1: TIEboard that provides exciting ways for Kids to learn complex shapes

ABSTRACT

This research is based on the concept of computing being embedded within the tangible product that acts as both input and output device eliminating the need of traditional computers for any feedback or guidance. The idea is inspired from traditional geoboard that focuses on the age group from 5 to 8 years old. The main goal is to integrate technology seamlessly into physical manipulative and while using this product kids will be able to make complex shapes that offer kids with memorable learning experience.

CCS CONCEPTS

• Human-centered computing → Interaction Design.

KEYWORDS

Tangible user interface (TUI), Tangible Interaction, Geometry, Kids Learning, Geoboard

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1 INTRODUCTION

The learning and advancement of mathematical knowledge and concepts is an important part of children's early academic development. Unfortunately, many children struggle with mathematics and study has indicated that geometry strand is the most neglected part in early education for kids [Moss J. 2015]. It has been indicated in the research that geometry learning manipulative have been evolving ever since—such as pattern blocks, tangram puzzles to Computed Aided games [Chi-Wei Lee 2008]. These physical manipulates help children develop better understanding of challenging concepts however, these manipulative do not provide any appropriate feedback to the kids to help in the learning process [Resnick et al. 1998]. With the rapid advancement in the technology many digital games have been introduced for kids mathematical learning, one such software game can be easily played on "splashlearn.com" that caters different levels of geometry [StudyPad 2020], however such interfaces lack in utilizing the richness of human senses and abilities. TUIs (tangible user interface), as opposed to the digital games, are designed to take advantage of the haptic interaction abilities [Zhou and Wang 2015]. In this research we propose to create mathematical toy TIEboard (Tangible Interface for Education) inspired from traditional geoboards to teach geometry that offers haptic feedback (Figure 1). The aim is to integrate physical manipulative with tangible user interface that provide relevant feedback to make their shape learning experience more memorable.

2 SYSTEM OF TIEBOARD

2.1 Design of Instruction

This research idea is inspired from traditional geoboards that are being used to explore geometry in early education. We aim to target the age group of 5-8 years old kids as the product will incorporate different levels from easy to complex to cater the needs of shape

learning during the early education. As a first basic idea, the user passes a string or rubber band through the pins according to the visual instructions on the board. The user selects the shape according to the level by pressing the mode switch. Then the location of the first pin to hang the string is visually indicated by a flickering LED. Then, after hanging the thread on the first pin, the LED for the next instruction will light up by pressing the button. The product will teach kids on different levels according to their skills that are based on Building blocks for little kids, a research based curriculum and will also improve their attention. Users can learn simple shapes, concepts such as similarity and symmetry, and even complex geometric shapes (Figure 2). In addition users can also make original shapes using ArduinoIDE or conductive sensing method.

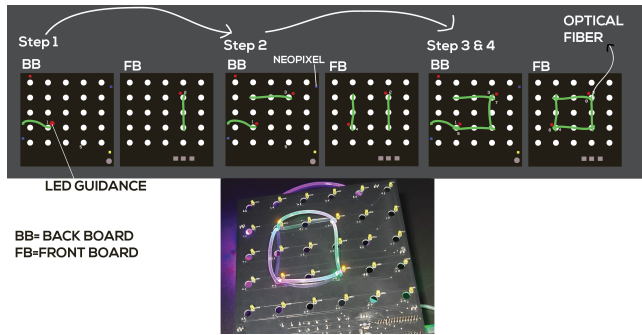


Figure 2: TIEboard: Step by Step Guidance to make a shape

2.2 Sensing Connection

When moving on to the following process, manually pressing a button might cause a mistake in the procedure. Therefore, we developed a method to sense the pin connection by using conductive rubber automatically. The resistance of the conductive rubber is 1.2×100 (Ohm.cm), and its diameter is 5mm. By controlling the matrix with eight horizontal pins over eight vertical pins, it is possible to sense which pins are connected to each other. For the microcomputer, we used an Arduino Nano. The aim was to give guidance through a red light, and when kids do correctly, it changes its color to green as positive feedback. It can also remember the original shape and share it with other users (Figure 3).

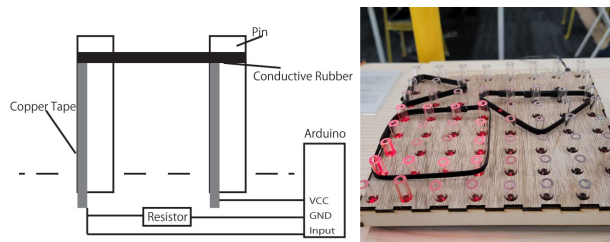


Figure 3: Sensing method with conductive rubber

2.3 Changing String Color

The color of the thread could not be changed dynamically in the normal geoboards. However in order to make the design more interactive and fun to play with we have introduced a method where kids can pass through the string (in this case optical fiber) to make shapes and along the way their string will glow with their choice of color and kids can even mix colors by controlling the neopixel LED's through potentiometer. In this case, we used an optical fiber (acrylic material) with a diameter of 3mm for illumination. The size of the holes are 7mm, and the number of holes is five vertically and six horizontally. LEDs were placed next to each of the holes on both sides of the board. While kids are lacing through to make shapes their shape learning is reinforced with colored optical fiber feedback. In this proposed design single color LED's are provided along the holes for appropriate guidance through variation in the blinking. After they finish passing through the hole, a button is pressed to get the next direction for the fiber to pass through. We also aim to introduce collaborative learning with the new design where kids can combine their boards and have more fun while make shapes and glowing their optical fiber together (Figure 4).



Figure 4: TIEboard: Shape learning is reinforced with colored optical fiber feedback and mixing colors by controlling neopixel

3 CONCLUSIONS AND FUTURE WORK

Research shows that physical manipulative plays a major role for kids to understand abstract mathematical concepts. In this research TIEboard will not only help kids to learn shapes but will also support them to develop problem solving skills. In addition, we would also like to research if this product will improve the hand-eye coordination in kids during early age of education.

For future, we would like to study if there can be the option given to upload any pattern for kids. We would also like to look into the possibility, if this can be used by any age group to write or make interesting things like neon signs for use in home, offices or study tables and make them glow in the choice of their favorite colors.

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